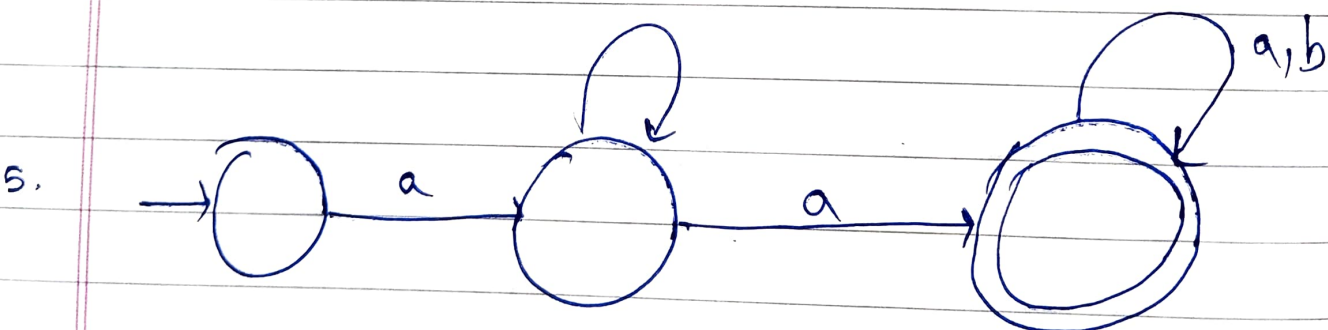
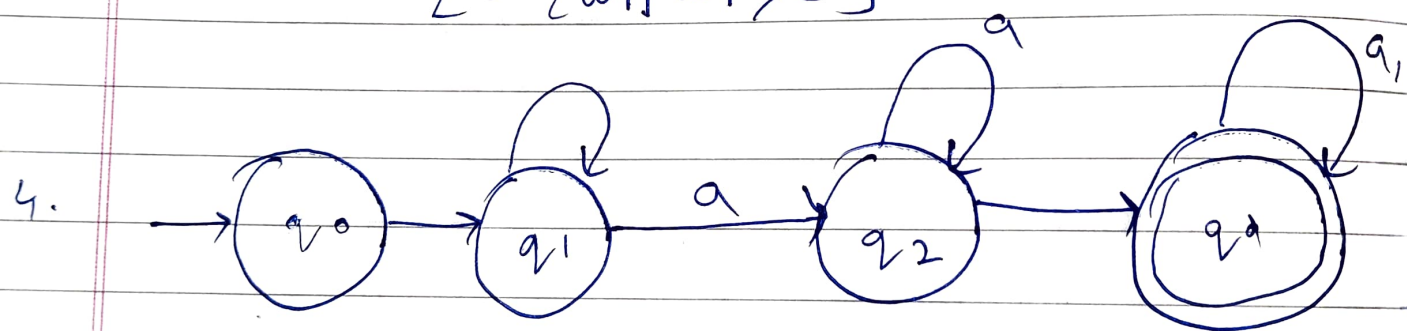
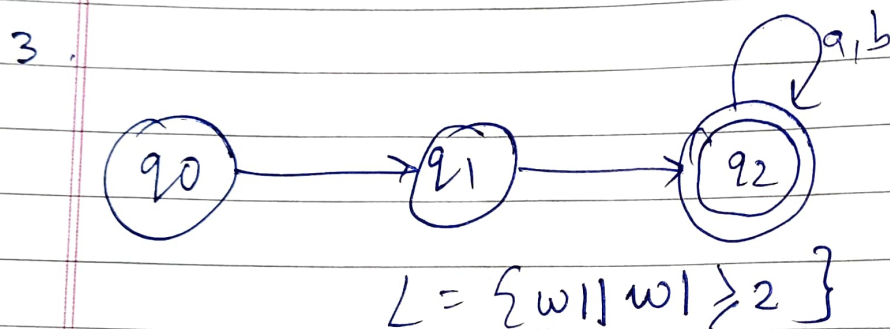
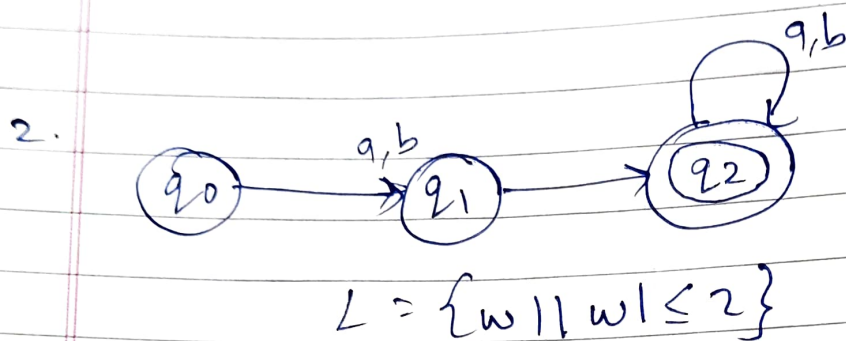
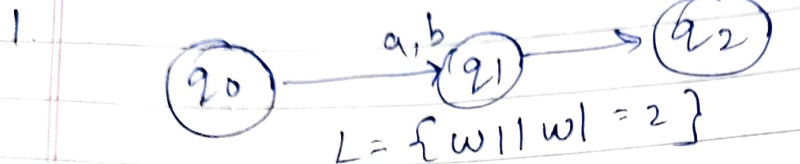


Problem 1



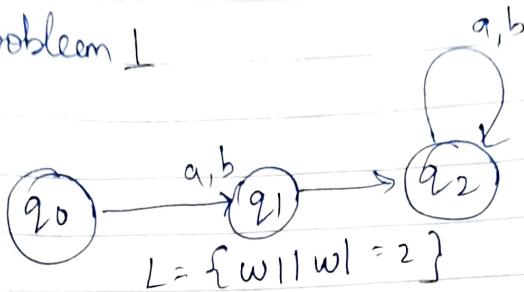
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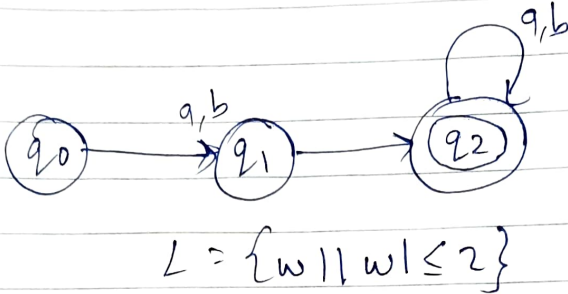
CSE-581

Problem 1

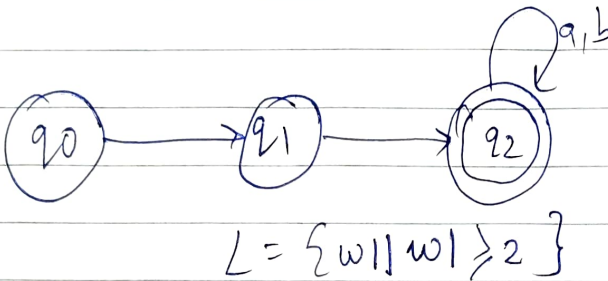
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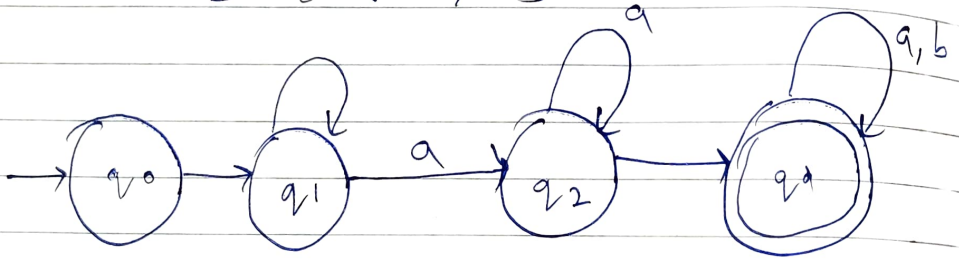
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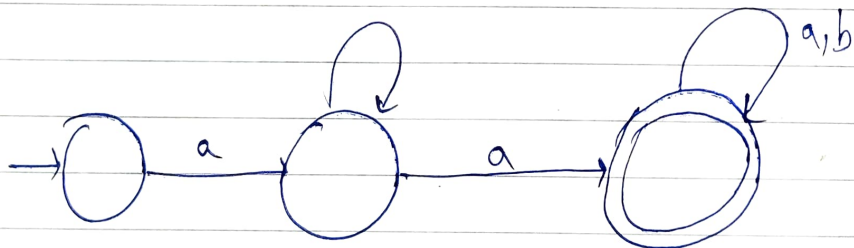
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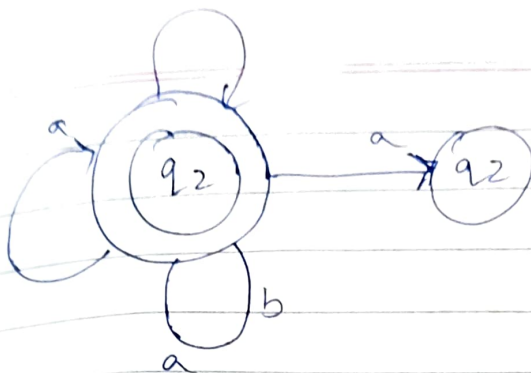
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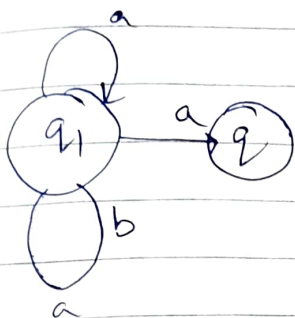
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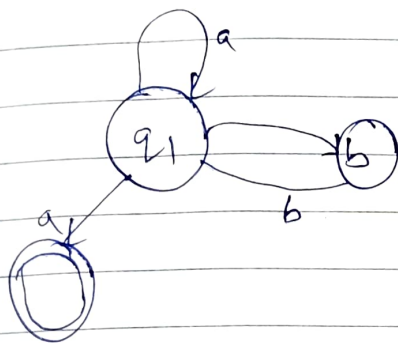
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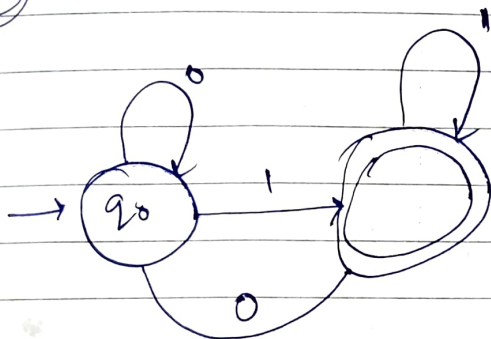
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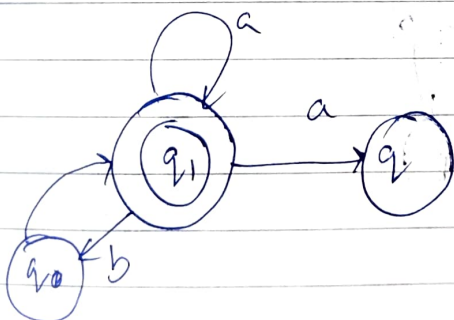
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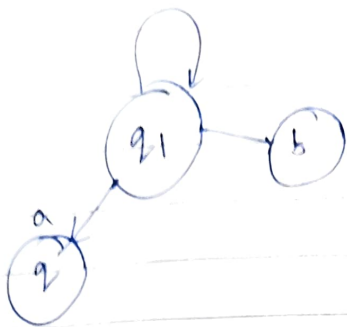
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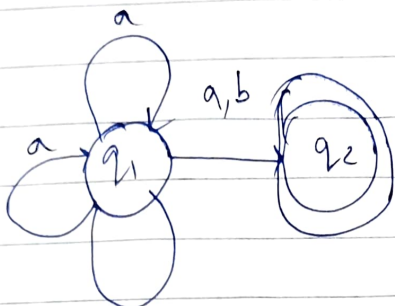
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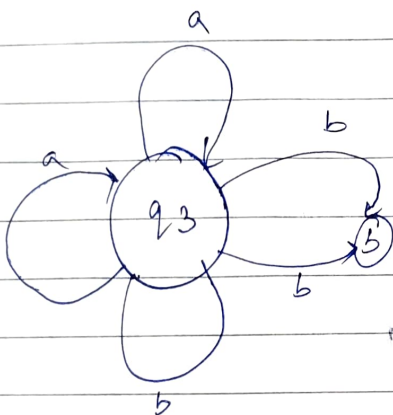
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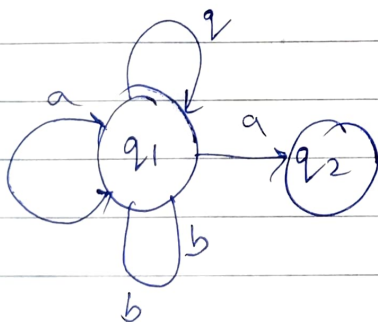
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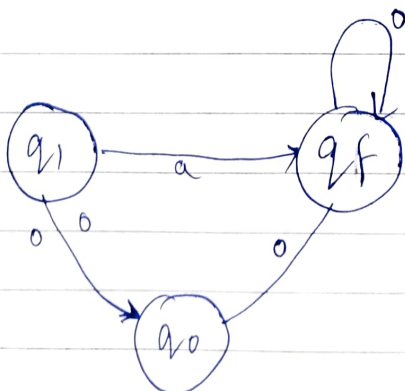
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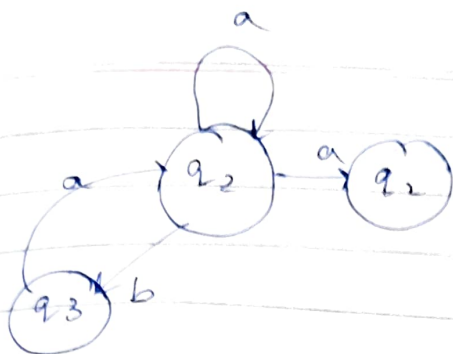
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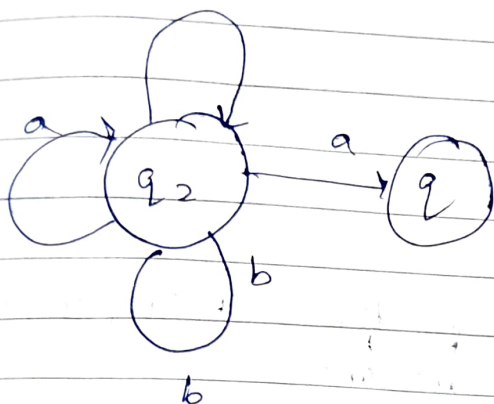
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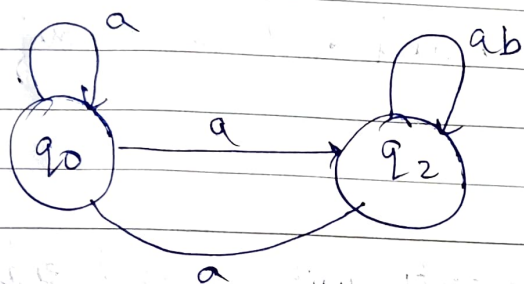
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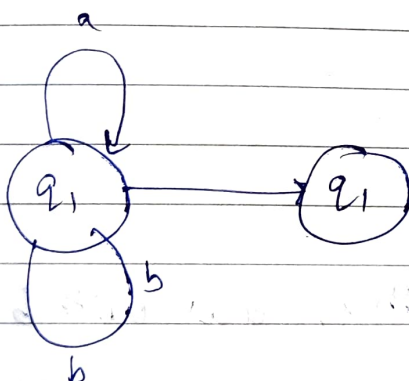
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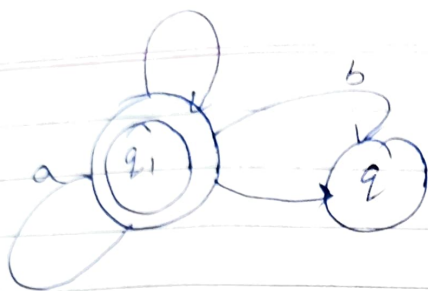


18.



19.





Q2 a) $L = \{w \mid w \text{ contains } ab\}$

$$(a+b)^*ab(a+b)^*$$

$(a+b)^*$ allows any prefix and suffix around ab must appear at least once

b) $L = \{w \mid w \text{ starts and ends with different symbols}\}$

start $\neq$ End

That means either starts with a and ends with b ,
or starts with b and ends with a .

string length ≥ 2

$$a(a+b)^*b + b(a+b)^*a$$

Explanation

$a(a+b)^*b$ starts with a , ends with b

$b(a+b)^*a$ starts with b , ends with a

choose string:

$$s = a^p a^p = ww^R \in L, \text{ where } w = a^p$$

So, $s = a^p a^p$, a palindrome.

let xyz be a split such that:

$$x = a^k$$

$$y = a^m, \text{ with } m > 0$$

$$z = a^{p-k-m} a^p$$

Now pump y :

$$xy^2z = a^{p+m} a^p \notin L \text{ since it is no longer symmetric}$$

× contradiction: $xy^2z \notin L$.

Thus, L is not regular.

2) $L = \{ \text{palindromes over } \{a, b\} \}$

claim: Not Regular

Assume L is regular

Pick $s = a^p b a^p \in L$, a palindrome of length $2p+1$

Let xyz be the pumping decomposition.

$$x = a^k, y = a^m, z = a^{p-k-m} b a^p$$

pumping $y \rightarrow xy^2z = a^{p+m} b a^p \notin L$.

x contradiction: Not a palindrome after pumping

Thus, not regular.

③ $L = \{a^n b^{2n} \mid n \geq 0\}$

claim: Not regular

choose $s = a^p b^{2p} \in L$

Let xyz be the decomposition where $|xy| \leq p$,
so all in a 's:

$$x = a^k, y = a^m, m > 0, z = a^{p-k-m} b^{2p}$$

Pump $y \rightarrow xy^2z = a^{p+m} b^{2p}$

Now, number of a 's $\neq$ half the number of b 's $\rightarrow$
Not in L .

x contradiction

Hence, not regular.

Q5
 (a) $L = \{wxw^R \mid x \in \Sigma^+, |w|, |x| \geq 1\}$

language is all strings of the form prefix + some middle string + reversal of prefix

This is a regular pattern

Let's consider

$$w \in \Sigma^+ \text{ (non-empty)}$$

$$x \in \Sigma^+ \text{ (non-empty)}$$

Defining

$$\text{Let } R_1 = \Sigma^+ \text{ (non-empty } w)$$

$$\text{Let } R_2 = \Sigma^+ \text{ (non-empty } x)$$

$$\text{Let } R_3 = \{w^R \mid w \in \Sigma^+\}$$

Now

$$L = \{wxw^R \mid w \in R_1, x \in R_2\}$$

Let $L_1 = \{wxw^R\}$. This is a regular transduction because

Palindromes are not regular, but here, reverse

happens only in a controlled, bounded part of the string.

~~to be~~

Since $|w| \geq 1$, for all pairs of (w, w^R) of bounded length L is regular

$$b) L = \{ xww^Ry \mid |x|, |y|, |w| \geq 1 \}$$

This is

prefix + palindromic segment + suffix

But the palindromic is only over ww^R , not the entire string.

we can construct a finite automata that

- verifies ww^R pattern exists somewhere inside
- Ensures that x and y at least length 1.

This is a regular expression with bounded conditions, and regular languages are closed under concatenation.

So,

$$R = \Sigma^+ \cdot \{ ww^R \mid w \in \Sigma^+ \} \cdot \Sigma^+$$

The center part is finite for fixed $|w|$
Thus, the language is regular.

Problem 3

NFA transition

states: $\{q_0, q_1\}$, start = q_0 , accept = q_0

Now

$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_0, b) = \{q_1\}$$

$$\delta(q_1, a) = \emptyset$$

$$\delta(q_1, b) = \{q_0\}$$

DFA states = subsets of $\{q_0, q_1\}$

DFA state contains q_0 on a on b Accept

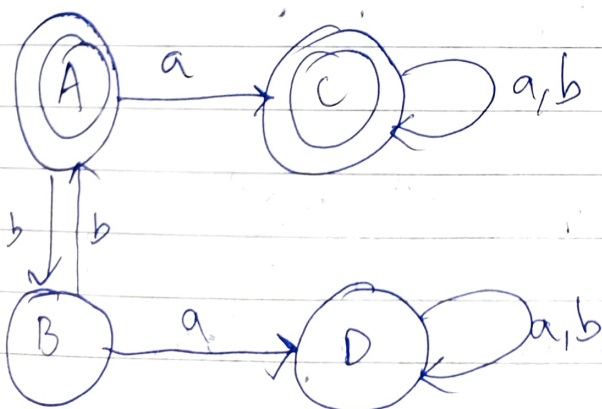
$\{q_0\}$ Yes $\{q_0, q_1\}$ $\{q_0, q_1\}$ Yes

$\{q_1\}$ No $\emptyset$ $\emptyset$ No

$\{q_0, q_1\}$ Yes $\{q_0, q_1\}$ $\{q_0, q_1\}$ Yes

$\emptyset$ Yes $\emptyset$ $\emptyset$ No

Let $A = \{q_0\}$, $B = \{q_1\}$, $C = \{q_0, q_1\}$



A, C are accepting states